



CHASR — Accounting Software Integration

Engineering Scope & Technical Specification

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1. Context & Objectives

Chasr is an automated accounts receivable (AR) platform for Australian SMEs that intelligently chases overdue invoices via configurable multi-channel sequences (email, SMS, and AI voice). The platform is live with 200+ customers, with a mature chase engine and Claude-powered messaging. Currently, invoice and debtor data is primarily populated through manual processes or limited imports, creating friction, data staleness, and adoption barriers.

Objective: Enable Chasr users to securely connect their existing accounting software (primarily Xero and MYOB) so that invoice, contact, and payment data is automatically ingested and continuously synced. This will populate the core Chasr data model (invoices, debtors, payment status), power accurate real-time chase workflows, eliminate manual data entry, reduce DSO through faster and more accurate interventions, and deliver a seamless experience consistent with the existing demo UI at the provided Replit link.

Strategic Note on Approach: Full native API integrations (OAuth 2.0 + webhooks where available) with Xero, MYOB and others remain the long-term target for real-time, low-friction, bi-directional capable sync. However, to deliver immediate value and accelerate platform population, a robust **mapped CSV/export workaround** (intelligent field mapping wizard, provider-specific templates, validation, and repeatable/scheduled uploads) will be built in parallel as the primary initial ingestion path. This mapped workaround normalises uploaded data into the exact same internal models used by the future API connectors, ensuring a seamless transition when API sync goes live.

The integration must support flexible authorization models including direct OAuth flows, secure credential provision, and invite-based delegation (e.g., to external accountants or bookkeepers) while maintaining the highest standards of security, privacy, and Australian regulatory compliance.

2. Connection & Authorization Methods

2.1 Primary: OAuth 2.0 Flow (Recommended)

Users authenticate directly with their accounting provider via the standard OAuth 2.0 Authorization Code flow (with PKCE for public clients). Chasr registers as an official app with Xero and MYOB developer programs. Minimal required scopes are requested (read-only: contacts, invoices, payments, organisations). Refresh tokens are securely stored and rotated. This method requires zero credential sharing with Chasr and provides granular consent and easy revocation.



2.2 Alternative: Secure Credential / API Key Provision

For scenarios where OAuth is unavailable or user prefers direct provision (e.g., legacy desktop MYOB instances or specific API key setups), the UI provides a secure form for entering username + password, API key + secret, or access token. All credentials are encrypted at rest using AES-256-GCM (key managed via environment/KMS), never logged in plaintext, and accessed only by the integration service layer under strict audit. Clear UI warnings and explanations of risks are mandatory. This mode is secondary and clearly labeled as less preferred.

2.3 Invite / Delegation Flow

Organisation admins can generate time-limited, single-use secure invite links or codes (cryptographically random, expiring in 7 days, rate-limited). These can be emailed or shared with authorised third parties such as external accountants or bookkeepers. The invitee visits a dedicated, branded connect page (outside full Chasr app if desired), completes OAuth or credential entry, and authorises the connection on behalf of the organisation. The token is single-use and provides no ongoing access to Chasr beyond the connection setup. This supports separation of duties and is especially useful when the primary user does not personally manage the accounting software login.

SECURITY NOTE: All methods enforce TLS 1.3 in transit. Stored tokens/credentials use envelope encryption. Every connection, sync, and access is immutably logged with actor, timestamp, action, and outcome for audit and breach detection. Connections can be revoked instantly from the UI, immediately invalidating tokens where possible.

3. Supported Accounting Platforms (MVP Focus)

Platform	Primary Auth	Sync Method	MVP Priority	Notes
Xero	OAuth 2.0	Webhooks + Polling	High — Phase 1	Excellent API, webhooks for invoices/contacts/payments. AU focus perfect match.
MYOB AccountRight / Essentials	OAuth 2.0 or API Key	Polling (webhooks limited)	High — Phase 1	Cloud-first; desktop instances may require credential or file-based fallback.
QuickBooks Online	OAuth 2.0	Webhooks + Polling	Medium — Phase 2	Strong API; add if client demand strong.
CSV / Manual Upload	N/A (file upload)	One-time or scheduled	High — Phase 1 (fallback)	Universal fallback + primary initial workaround. Smart field mapping wizard, pre-built templates for common Xero/MYOB/QuickBooks exports, validation, and repeatable uploads. Bridges to full API sync.



4. Data Ingestion Scope & Mapping

The integration performs an initial bulk import on connection (with progress indicator and estimated time), followed by ongoing incremental synchronisation. Accounting system is treated as the source of truth for core invoice and payment data. Chasr augments with chase-specific metadata (chase_state, communications log, etc.).

4.1 Core Entities to Ingest

Entity	Key Fields (Minimum)	Purpose in Chasr
Contacts / Debtors	External ID, Name, Email(s), Phone(s), Address, Payment Terms, Notes	Debtor profiles, chase personalisation, contact matching for voice/email/SMS
Invoices	External ID, Invoice #, Date, Due Date, Total, Balance Due, Status, Contact ID, Currency, PDF URL (if avail)	Core chase queue, aging reports, sequence triggering, amount accuracy
Payments / Allocations	External ID, Date, Amount, Allocated Invoice(s), Method, Reference	Auto stop chase on full/partial payment, accurate balance tracking, forecasting input
Credit Notes / Adjustments (Phase 2)	External ID, Linked Invoice, Amount, Reason, Date	Correct balances, prevent erroneous chasing

Initial sync targets active + recently closed invoices (configurable lookback, default 18 months) and all current contacts with open items. Historical closed invoices can be optionally included for analytics/forecasting in later phases.



4.2 Initial Mapped Workaround — CSV/Export Ingestion with Intelligent Field Mapping

While native API connectors are under development, Chasr will ship a first-class **mapped CSV/export ingestion module** as the primary initial data population mechanism. This provides immediate utility for all users regardless of accounting platform and serves as the technical foundation that the later API sync layer will reuse (same normalisation, deduplication, and chase-enrolment logic).

Key capabilities of the mapped workaround:

- Pre-built, downloadable CSV/Excel templates aligned to common Xero, MYOB AccountRight, QuickBooks, and generic AR ledger exports (columns for Invoice #, Date, Due Date, Amount, Balance, Contact Name/Email/Phone, Status, etc.).
- Interactive **field mapping wizard** in the UI: Users upload a sample file → system auto-detects columns where possible → user confirms or adjusts mappings via dropdowns or drag-and-drop. Mappings can be saved as named templates per organisation or shared.
- Robust parsing & validation: Automatic date/amount/currency normalisation (AU formats), duplicate detection (by external ID or fuzzy matching on invoice # + contact), required field checks, and clear error reporting with row-level highlighting.
- One-time or recurring uploads: Support for manual drag-and-drop plus optional scheduled SFTP/email-drop or recurring 'upload reminder' workflows for teams that export weekly from their accounting system.
- Post-upload processing: Data flows through the identical ingestion pipeline as future API sync (normalisation → upsert by external ID → balance/status updates → chase engine triggers for new overdue items and payment halts).
- Audit & rollback: Every upload creates a sync log entry with record counts, errors, and the ability to preview changes before committing or to roll back a batch if needed.

This approach ensures that when the Xero/MYOB OAuth + webhook connectors are activated, existing users experience zero data migration effort — the platform simply switches the data source while preserving all chase history and configurations.

5. Sync Architecture & Technical Approach

High-level components: Connector layer (XeroConnector, MyobConnector — using official SDKs where available), SyncOrchestrator (BullMQ / agenda / custom queue for jobs), WebhookReceiver (Express endpoints for provider push), DataMapper (normalisation + conflict resolution), AuditLogger, and CredentialVault (encrypted storage abstraction).

Sync triggers: (1) Manual 'Sync Now' from UI, (2) Scheduled polling jobs (configurable per connection, default 30 min), (3) Incoming webhooks from provider (preferred for Xero — real-time invoice update, payment, contact change). All jobs are idempotent using (org_id + external_id) as natural key. Large datasets (>5k records) use cursor-based pagination and background processing with UI progress polling.

On payment sync: If invoice balance_due becomes zero or status changes to PAID, immediately update Chasr invoice record and signal the chase engine to halt any active sequence for that invoice. New invoices that meet 'overdue or due within N days' criteria are automatically enrolled into the client's configured chase sequence.

Conflict policy (MVP): Accounting system wins for core financial fields (amounts, dates, status). Chasr-specific fields (chase_state, last_chase_at, communications) are preserved or merged. Future phases may add manual override UI.



6. User Interface & Experience Requirements

New top-level navigation item or prominent Settings section: **Integrations** (or 'Accounting Connections'). Consistent visual language with the existing Chasr demo (clean cards, status pills, professional tables, teal/navy accents).

Key Screens / Flows:

- **Integrations Dashboard:** Grid or list of connected sources with logo, connection name, status pill (Connected / Syncing / Error / Disconnected), last sync time, record counts (e.g. '1,284 invoices'), 'Sync Now' and 'Manage' actions.
- **Add Connection Flow:** Modal or dedicated page with provider selection cards (Xero prominent, MYOB, CSV Upload, 'Other / Invite Accountant'). Clicking Xero/MYOB launches OAuth redirect. CSV Upload opens the full **smart mapping wizard** (template download, auto column detection, interactive field mapping with saveable templates, validation preview, and commit). This is the primary initial population path.
- **Invite Accountant Flow:** Button generates unique secure link + optional email. Link lands on a lightweight branded page explaining the request and collecting OAuth/credentials. Success message notifies original admin.
- **Connection Detail / Manage:** Sync history log (last 50 runs with duration, records processed, errors), re-authorise button, disconnect (with confirmation and data retention note), advanced settings (sync frequency, lookback period, field overrides if Phase 2).
- **Onboarding / First Run:** Enhanced new organisation wizard includes prominent 'Connect your accounting software' step with benefits callout (auto population, real-time payments, reduced manual work).
- **Dashboard Integration:** Subtle badge or widget on main invoice list / overview: 'Data synced from Xero • Last update: 4m ago' with link to Integrations. Data freshness indicator or warning if sync stale > X hours.
- **Error & Notification Handling:** In-app banners and email alerts for sync failures, re-auth required (token expiry/revoke), rate limit warnings. Clear actionable messages (e.g. 'Xero connection needs re-authorisation — click here').

All flows must be mobile-responsive and accessible (WCAG AA target). Loading states, skeleton screens, and optimistic UI updates for better perceived performance during initial bulk syncs (which may take minutes for large ledgers).



7. Security, Privacy & Compliance

This feature handles sensitive financial data and authentication credentials/tokens. Security is non-negotiable and must be designed in from the start, not bolted on. The implementation will undergo internal security review and preferably external penetration testing prior to production release.

- **Encryption:** All tokens and credentials encrypted at rest with AES-256-GCM. Encryption keys managed separately (KMS or dedicated secrets service). In transit: TLS 1.3 minimum.
- **Least Privilege:** OAuth scopes limited to read-only invoice, contact, payment, and organisation data. No write access requested in MVP.
- **Audit & Logging:** Immutable append-only logs for every connection creation/revoke, sync start/complete/fail, credential access, and data mutation. Logs include actor (user or system job), IP if applicable, and outcome.
- **Data Minimisation & Retention:** Only ingest fields required for chase functionality + basic reporting. Configurable retention for raw payloads (default 90 days) and normalised records aligned to active chase + statutory requirements.
- **Privacy Act Compliance:** Clear consent language in UI explaining exactly what data is accessed and why. APP 1 (open & transparent), APP 3 (collection), APP 6 (use/disclosure), APP 11 (security) addressed. Ready for Notifiable Data Breach response.
- **Invite Security:** Cryptographically secure random tokens (min 256-bit entropy), single-use, short expiry (7 days), rate-limited generation and redemption, optional IP or email domain binding.
- **Revocation:** Instant disconnect from UI invalidates Chasr-side tokens and notifies user of provider-side revocation steps where relevant.
- **Multi-Tenancy:** Strict row-level security and org-scoping in all queries. No cross-organisation data leakage possible.

8. Recommended Phased Implementation

Phase	Scope	Key Deliverables	Est. Duration
Phase 1 MVP	Primary focus: Robust mapped CSV/export workaround as immediate population path (smart mapping wizard, templates, validation, repeatable uploads) + foundational Xero OAuth stub. Unified ingestion pipeline, stop-on-payment, core Integrations UI + flows. Basic credential mode as stretch.	Mapped CSV engine + smart mapping wizard + templates, unified normaliser, DB migrations, core Integrations UI + flows (CSV primary + Xero OAuth stub), audit logging, basic sandbox validation. Documentation & handover.	2 weeks (focused MVP)
Phase 2 Enhance	MYOB full support (OAuth priority). Real-time webhooks for Xero. Invite/delegation flow. Advanced mapping & conflict UI. Detailed sync logs & export. Performance tuning for 10k+ invoice orgs. Onboarding integration.	MYOB connector, webhook receiver + handlers, invite token service + UI, sync history viewer, field mapping admin, email notifications, load testing report.	Additional ~4 weeks
Phase 3 Future	QuickBooks Online, bi-directional sync (chase notes → invoice comments), deeper financial data pull for forecasting, configurable per-client mappings, white-label accountant portal, advanced analytics from accounting data.	Additional connectors, write-back service, forecasting module hooks, admin mapping UI, multi-tenant accountant features.	TBD — backlog



9. Deliverables & Acceptance Criteria

- Approved engineering scope document (this PDF) and any supporting technical design diagrams.
- Production-ready, tested code for all Phase 1 components committed to main branch with feature flags.
- Database migrations and seed data for new tables/columns.
- Frontend components and pages integrated into existing Chasr UI (matching demo styling and UX patterns).
- Comprehensive internal documentation: runbooks for connectors, troubleshooting sync issues, data mapping reference.
- User-facing help articles / in-app tooltips for connection flows (copy provided or drafted).
- Sandbox + staging validation report using real Xero/MYOB test companies (minimum 3 diverse datasets).
- Security review sign-off and basic penetration test results (or plan if external).
- Post-launch hypercare support window (minimum 10 business days) including monitoring and rapid fixes.

Acceptance Criteria (Phase 1):

A typical SME user can complete Xero OAuth connection in under 3 minutes from Integrations page. First invoices and contacts appear in the main Chasr dashboard within 10 minutes of connection. Subsequent payment updates from Xero stop active chases within 5 minutes (or immediately via webhook). No plaintext credentials stored. All sync errors surface clearly in UI with actionable guidance. 95%+ field mapping accuracy on test datasets. Mobile flows functional. Audit logs capture 100% of connection and sync events.

10. Assumptions, Dependencies & Risks

Key Assumptions:

- Access to full existing Chasr codebase (api-server, frontend, DB schema, chase engine, auth system).
- Ability to register Chasr as official apps in Xero and MYOB developer portals (developer accounts provided).
- Sandbox/test company access in Xero and MYOB for validation (or ability to create realistic synthetic data).
- Existing frontend framework supports the new UI patterns (React/TS assumed from demo context).
- Queue/job processing infrastructure (BullMQ or equivalent) already in use or acceptable to introduce.
- User/organisation multi-tenancy model already established and documented.

Identified Risks & Mitigations:

Risk	Impact	Mitigation
API changes or deprecations by Xero/MYOB	High — breaks sync	Use official SDKs, version pinning, abstraction layer, monitoring for deprecation notices, fallback polling.
Large ledgers cause slow initial sync / timeouts	Medium — poor UX	Background jobs + progress polling, cursor pagination, incremental first sync option, performance testing early.
User confusion on OAuth vs credential vs invite	Medium — support load	Clear UI copy, tooltips, 'Recommended' badges, short video or animated guide, progressive disclosure.



Token/credential leakage or breach	Critical — compliance & trust	Encryption by default, minimal storage time, immediate revocation UI, audit logging, incident response plan.
Data mapping edge cases (tax, multi-currency, custom fields)	Medium — data quality	Conservative MVP mapping, extensive test cases, 'report issue' button that captures raw payload for support.

This scope provides a clear, actionable blueprint for delivering secure, user-friendly accounting integrations that will significantly increase Chasr's value proposition, reduce onboarding friction, and improve data accuracy for the chase engine. It aligns with Chasr's existing engineering standards (as seen in the AI Voice System Engineering Brief) and the live demo UI experience.

Next steps: Review and approve this scope (or provide feedback for iteration), confirm Phase 1 timeline and resource allocation, provision developer accounts and codebase access, and schedule kick-off for technical design deep-dive if required.

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Prepared by OmniSuiteAI on behalf of the Chasr platform team. For questions or clarifications on this scope, contact the project lead or engineering team.